



232. Implement Queue using Stacks

Solved

Easy Topics

Implement a first in first out (FIFO) queue using only two stacks. The implemented queue should support all the functions of a normal queue (`push`, `peek`, `pop`, and `empty`).

Implement the `MyQueue` class:

- `void push(int x)` Pushes element `x` to the back of the queue.
- `int pop()` Removes the element from the front of the queue and returns it.
- `int peek()` Returns the element at the front of the queue.
- `boolean empty()` Returns `true` if the queue is empty, `false` otherwise.

Notes:

- You must use **only** standard operations of a stack, which means only `push` to `top`, `peek/pop` from `top`, `size`, and `is empty` operations are valid.
- Depending on your language, the stack may not be supported natively. You may simulate a stack using a list or deque (double-ended queue) as long as you use only a stack's standard operations.

Example 1:

Input

```
["MyQueue", "push", "push", "peek", "pop", "empty"]
```

</> Code

Python3 Auto

```

1 class MyQueue:
2
3     def __init__(self):
4         self.s1=[]
5         self.s2=[]
6
7     def push(self, x: int) -> None:
8         self.s1.append(x)
9
10    def pop(self) -> int:
11        self.peak()
12        return self.s2.pop()
13
14    def peek(self) -> int:
15        array_length = len(self.s2)
16        if array_length == 0:
17            while len(self.s1) != 0:
18                self.s2.append(self.s1.pop())
19        return self.s2[-1]
```

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 [>_ Test Result](#)

You must